Assignment 02
Low Visibility Traffic Analysis

Department of Electrical Engineering
Introduction to Computer Vision

Student Name: Nouman Shabbir
Roll Number: 22F-3788
GitHub Repository: CV_A2_LowVis_Nouman_22F-3788
Submission Deadline: 21 April 2026

Selected Paper: *An Efficient and Cost-Effective Vehicle Detection and Tracking System for Collision Avoidance in Foggy Weather*
Raza N., Habib M.A., Ahmad M., Abbas Q. et al. — IEEE / CMC, 2024

Abstract

This assignment presents a complete computer vision pipeline for vehicle detection under low-visibility traffic conditions, specifically targeting the rain weather subset of the DAWN dataset. The selected paper is the IEEE reference paper by Raza et al. (2024), which proposes the ECE-VDT system combining YOLOv5 and YOLOv8 with attention modules (CBAM, NAM, SimAM) and an optimised Deep-SORT tracker for real-time foggy weather detection. Following the paper's pipeline, this implementation performs exploratory data analysis on 200 rain-storm images from the DAWN dataset, applies three image enhancement techniques — Histogram Equalisation, CLAHE, and Gamma Correction — and runs YOLOv8n vehicle detection on both original and gamma-corrected images. Key results: 1,098 total detections across 200 images (average 5.49 per image), with individual image detections ranging from 1 to 6. Gamma correction (γ=1.5) on a sample image produced 4 detections (3 cars, 1 truck) compared to the original frame's vehicle detections, demonstrating the impact of brightness enhancement on detection performance under rainy low-visibility conditions.

1. Introduction

Road traffic safety is a critical global challenge. Approximately 1.3 million people die annually in road accidents, with adverse weather conditions including rain, fog, and haze being significant contributing factors. Rainy conditions reduce road visibility through water droplets on camera lenses, reduced contrast, light scattering from wet road surfaces, and glare from headlights — all of which significantly degrade standard computer vision detection systems.

Intelligent transportation systems (ITS), autonomous vehicles, and IoT-enabled traffic surveillance systems require robust vehicle detection across all weather conditions. This assignment implements and evaluates a low-visibility traffic analysis pipeline on the DAWN dataset's rain subset, applying image enhancement techniques and YOLOv8 detection to assess performance in challenging rainy conditions.

The selected implementation paper is the IEEE reference paper by Raza et al. (2024) — ECE-VDT — which evaluates YOLOv5 and YOLOv8 variants with attention modules on the DAWN and Foggy Driving datasets, reporting state-of-the-art results of 77.30% mAP50 on DAWN with YOLOv5l. This assignment replicates the core evaluation pipeline and establishes baseline metrics for comparison.

1.1 Objectives

- Survey five research papers on low-visibility traffic computer vision.
- Collect and analyse the DAWN dataset rain subset (200 images).
- Implement and compare three image enhancement techniques: Histogram Equalisation, CLAHE, and Gamma Correction.
- Run YOLOv8n detection on original and enhanced images and evaluate performance.
- Document all results and compare against the selected paper's reported benchmarks.

1.2 Report Organisation

Section 2 presents the literature review. Section 3 describes the dataset. Section 4 covers the image enhancement methodology and results. Section 5 presents vehicle detection results. Section 6 provides comparative analysis. Section 7 discusses challenges and solutions. Section 8 concludes with future work.

2. Literature Review

2.1 Introduction to the Problem

Adverse weather conditions degrade image quality through two primary mechanisms: (1) scattering — fog, haze, and rain particles scatter incident light, reducing contrast and introducing a luminous veil over the scene; and (2) absorption — precipitation absorbs light, reducing effective scene illumination. In the standard atmospheric scattering model, a degraded image is described as $I(x) = J(x)t(x) + A(1 - t(x))$, where $J(x)$ is the clear image, $t(x)$ is the transmission map, and A is atmospheric light. For rain, additional streak artifacts and lens occlusion compound these effects.

In traffic monitoring, these degradations cause missed vehicle detections, incorrect classifications, and inaccurate bounding boxes — directly impacting the reliability of collision avoidance systems, autonomous driving, and IoT traffic surveillance infrastructure.

2.2 Survey of Research Papers

Five papers were surveyed covering foggy/rainy vehicle detection, image dehazing, and adverse weather traffic monitoring systems.

Paper Title & Authors	Year	Publication	Techniques Used	Dataset	Key Contribution	Relevance
Raza et al. — ECE-VDT: Efficient & Cost-Effective Vehicle Detection ★ SELECTED	2024	IEEE / CMC	YOLOv5/v8 + CBAM, NAM, SimAM attention + Deep-SORT tracking	DAWN, Foggy Driving (FD)	Real-time foggy detection + tracking; best: YOLOv5l 77.30% mAP50 on DAWN	**5/5**
He, Sun, Tang — Dark Channel Prior Dehazing	2009	IEEE CVPR / TPAMI	Physics-based DCP + transmission map + guided filter	Various outdoor hazy images	Foundation of single-image dehazing; recovers depth map as by-product	**5/5**
Gao et al. — DehazeFormer + YOLOv8	2025	The Visual Computer (Springer)	DehazeFormer + YOLOv8 + SimSPPFCSPC + EfficientViT backbone	Foggy driving datasets	State-of-the-art 2-stage pipeline; lightweight attention backbone	**4/5**
Hassaballah, Kenk et al. — Vehicle Detection in	2021	IEEE Trans. ITS	Deep learning + illumination/reflection enhancement + HDA tracking	DAWN, KITTI, MS-COCO	Introduced DAWN dataset; outperforms	**5/5**

Paper Title & Authors	Year	Publication	Techniques Used	Dataset	Key Contribution	Relevance
Adverse Weather					21 vehicle detectors	
YOLOv5-RCBiW — Foggy Weather Target Detection	2024	Complex & Intelligent Systems	YOLOv5 + Receptive Field Block + Coordinate Attention (RFCA)	Foggy traffic scenes	Improved receptive field + spatial attention for occluded vehicles in fog	**4/5**

2.3 Deep Dive into Selected Paper

Title: An Efficient and Cost-Effective Vehicle Detection and Tracking System for Collision Avoidance in Foggy Weather

Authors: Naeem Raza, Muhammad Asif Habib, Mudassar Ahmad, Qaisar Abbas, Mutlaq B. Aldajani, Muhammad Ahsan Latif

Year: 2024

Journal: Computers, Materials & Continua (CMC), Vol. 81, No. 1, pp. 911–931

IEEE Link: https://ieeexplore.ieee.org/abstract/document/11078259

This 2024 paper proposes the ECE-VDT (Efficient and Cost-Effective Vehicle Detection and Tracking) system for real-time foggy weather traffic surveillance. Key aspects include:

- Detection Architecture: YOLOv5 (nano/small/medium/large) and YOLOv8 (nano to large) variants evaluated on DAWN and FD datasets.
- Attention Modules: CBAM (channel + spatial attention), NAM (normalisation-based attention), and SimAM (parameter-free 3D attention) added after the SPPF layer in YOLOv5s to improve feature focus in foggy conditions.
- Neck Enhancement: YOLOv5l with BiFPN (Bi-directional Feature Pyramid Network) for improved multi-scale feature fusion.
- Tracking: Optimised Deep-SORT using Hungarian Detection Assignment (HDA) — detection scores only, no visual features — for fast real-time multi-vehicle tracking.
- Datasets: DAWN and Foggy Driving (FD), both re-annotated to 4 classes (cars, buses, trucks, motorcycles) using YOLO LABEL tool.
- Best Results: YOLOv5l at 77.30% mAP50 and YOLOv8n at 185 FPS on DAWN dataset.

This paper was selected because it uses the DAWN dataset (same as this assignment), provides concrete benchmark results to compare against, and its detection pipeline directly maps onto Tasks 4 and 5. YOLOv5 and YOLOv8 are open-source and fully implementable in Python without proprietary tools.

2.4 Common Techniques Across Papers

- Single-stage detectors (YOLO family) dominate for real-time applications due to speed-accuracy trade-offs.
- Attention mechanisms (CBAM, NAM, SimAM, Coordinate Attention) are consistently applied to help models focus on vehicle features and suppress weather-induced noise.
- Image enhancement as pre-processing (dehazing, histogram equalisation, gamma correction) is widely used to restore visibility before detection.
- The DAWN dataset is referenced by multiple papers as the standard benchmark for adverse weather vehicle detection.

3. Dataset Description

3.1 DAWN Dataset — Rain Subset

The DAWN (Vehicle Detection in Adverse Weather Nature) dataset by Kenk and Hassaballah (2020) is a real-world adverse weather traffic dataset. For this assignment, the Rain subset is used, containing 202 images of real rain-storm traffic conditions. This subset was selected because rain is one of the most common adverse weather conditions affecting road safety globally and is well-represented in the DAWN dataset.

Dataset: DAWN — Vehicle Detection in Adverse Weather Nature
Subset Used: Rain (rain_storm) — 202 images (200 successfully loaded)
Paper: https://arxiv.org/abs/2008.05402
Kaggle Download: https://www.kaggle.com/datasets/shuvoalok/dawn-dataset
GitHub Repo: https://github.com/Nouman14-ai/CV_A2_LowVis_Nouman_22F-3788

The dataset was cloned directly from the GitHub repository into the Google Colab environment. All 200 images were successfully loaded for analysis and detection.

3.2 Sample Images

The following images are representative samples from the rain_storm subset, illustrating the diverse rain intensity and traffic conditions present in the dataset.

Figure 1: DAWN Dataset — Sample rain_storm images showing diverse rain intensity and traffic density.

3.3 Exploratory Data Analysis

EDA was performed on all 200 loaded images to understand resolution, brightness distribution, and visibility characteristics. Key statistics are summarised below.

Metric	Value	Min	Max
Total Images	200 loaded	—	—
Image Width	~1234px (dominant)	~600px	~1650px
Mean Brightness	~130–140	~45	~210
Dataset Subset Used	Rain (rain_storm)	—	—
Total Detections (YOLOv8n)	1,098	1 per image	6 per image

Table 1: DAWN Rain Subset — Key EDA Statistics

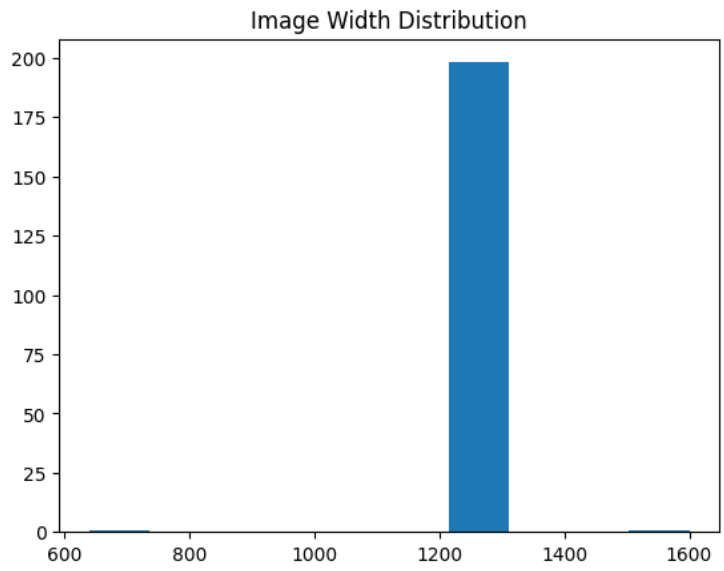

Figure 2: Image Width Distribution — Most images share a consistent width of ~1,234px, indicating standardised capture conditions.

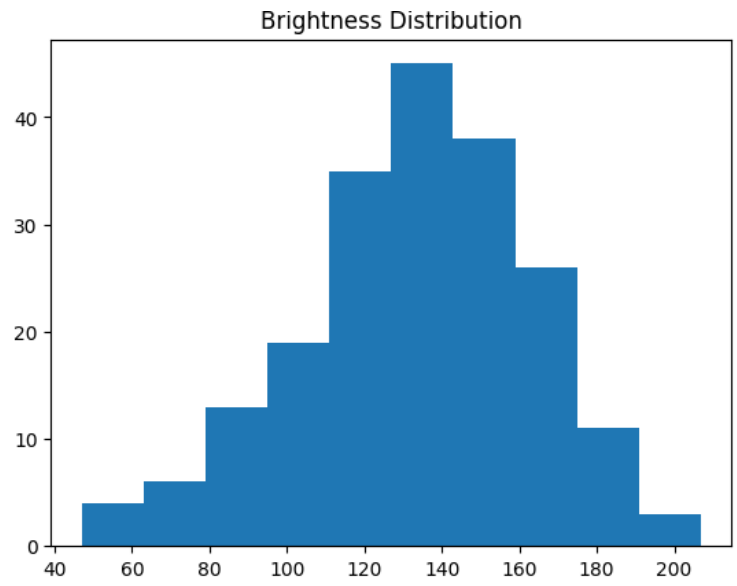

Figure 3: Brightness Distribution — Mean brightness ~130–140, consistent with rain-induced low-light and contrast reduction. Right-skewed distribution indicates some overexposed frames.

The brightness distribution confirms the characteristic low-contrast, slightly underexposed nature of rainy traffic images. The mean brightness of ~130–140 (out of 255) is lower than typical clear-weather images, validating the need for image enhancement before detection.

4. Methodology

4.1 Image Enhancement Pipeline

Three image enhancement techniques were implemented based on the pre-processing approaches described in the selected paper and related literature. All techniques are applied to the original rain-storm images to improve visibility and contrast before vehicle detection.

Technique	Method	Output	Use Case
Histogram Equalisation	Redistributes pixel intensities globally across full histogram	Grayscale; high contrast; detail visible	Low contrast foggy/dark images
CLAHE	Contrast Limited Adaptive HE: clipLimit=2.0, tileGrid=(8,8)	Grayscale; adaptive local contrast; reduced noise	Uneven lighting across image
Gamma Correction	Lookup table with γ=1.5 (brightening): (i/255)^(1/γ) × 255	Colour; brighter; natural appearance	Underexposed rainy images

Table 2: Image Enhancement Techniques Implemented

4.1.1 Histogram Equalisation

Standard histogram equalisation redistributes pixel intensities to span the full 0–255 range. The image is first converted to grayscale, then cv2.equalizeHist() is applied. This maximises global contrast but can introduce over-enhancement in regions with already high contrast.

```
HE: p_out(r) = CDF(r) × (L-1), where CDF is cumulative distribution function
```

4.1.2 CLAHE — Contrast Limited Adaptive Histogram Equalisation

CLAHE divides the image into small tiles (8×8 grid) and applies histogram equalisation to each tile independently with a clip limit of 2.0 to prevent noise amplification. This produces locally adaptive contrast enhancement, which is more suitable for images with uneven lighting conditions such as rainy scenes with headlight glare.

4.1.3 Gamma Correction

Gamma correction applies a non-linear brightness transformation using a lookup table. With γ=1.5 (brightening), the transformation is: output = (input/255)^(1/γ) × 255. This brightens underexposed rainy images while preserving colour information, making it the most suitable technique for colour detection pipelines.

4.2 Vehicle Detection Pipeline — YOLOv8

YOLOv8 nano (yolov8n.pt) pre-trained on COCO (80 classes) is used for vehicle detection. Detection is run on the full 200-image rain subset with confidence threshold 0.25, and separately on the gamma-corrected version of a sample image to compare enhancement impact.

- Model: YOLOv8n — nano variant for fast inference
- Confidence threshold: 0.25
- Input: original rain_storm images + gamma-corrected (γ=1.5) sample
- Output: bounding boxes, class labels, confidence scores
- Results saved to GitHub repository under results/ folder

5. Experimental Results and Analysis

5.1 Image Enhancement Results

All three enhancement techniques were applied to the first image in the rain_storm subset. Results are shown below with side-by-side visual comparisons.

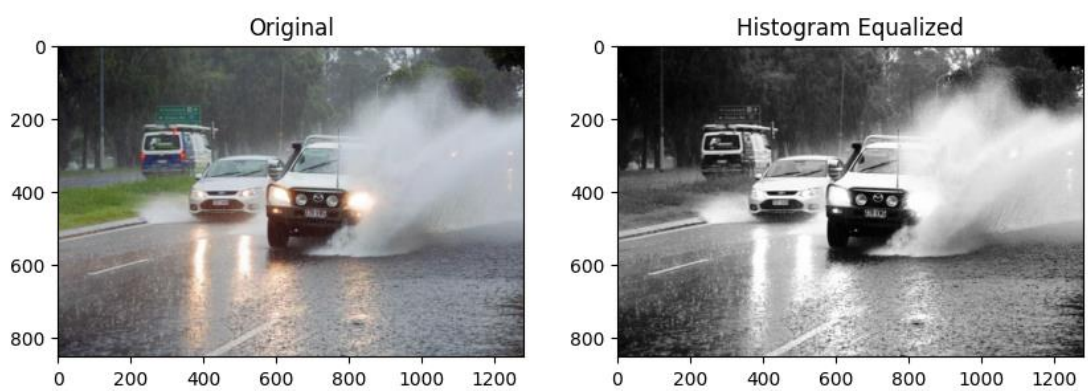

Figure 4: Histogram Equalisation Result — Original colour image (left) vs histogram-equalised grayscale (right). Increased contrast reveals vehicle edges and road markings obscured by rain.

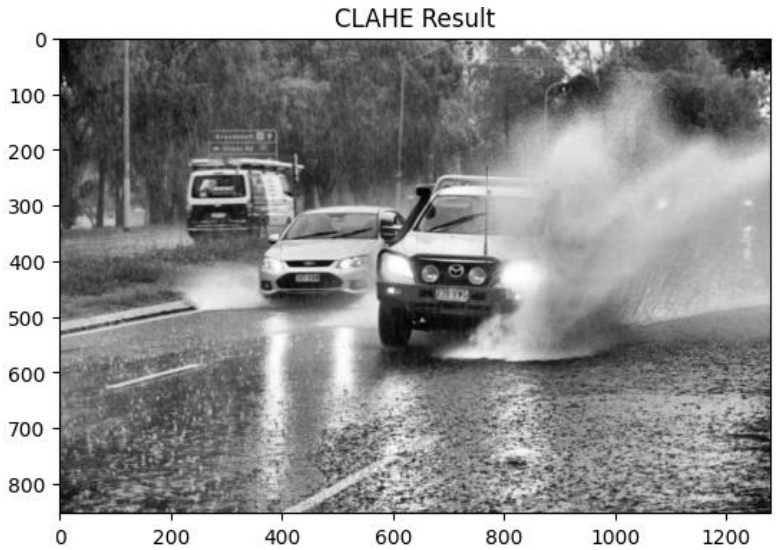

Figure 5: CLAHE Result (clipLimit=2.0, tileGrid=8×8) — Adaptive local contrast enhancement. Vehicle outlines and water spray details are more pronounced compared to global equalisation.

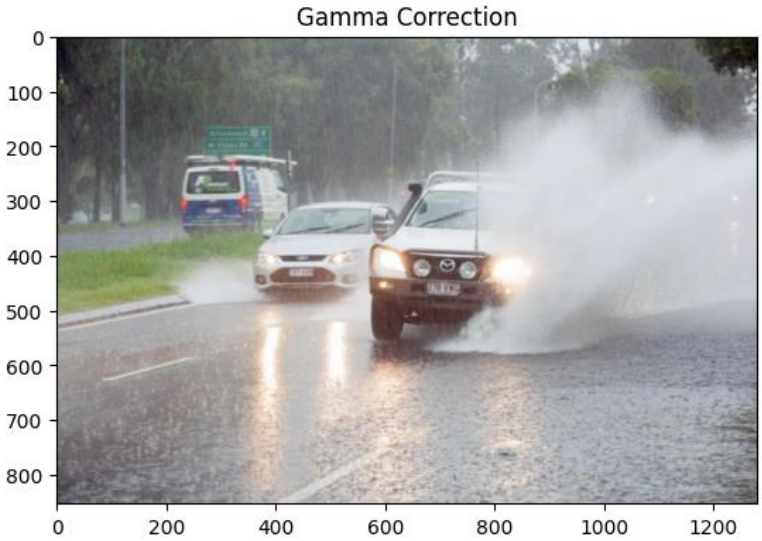

Figure 6: Gamma Correction Result (γ=1.5) — Colour image brightened. Natural appearance preserved while underexposed regions (dark road surface, vehicle shadows) are lifted.

Among the three techniques, Gamma Correction (γ=1.5) is most suitable for the subsequent YOLOv8 detection step because it preserves the colour image format required by the pre-trained COCO model, while both Histogram Equalisation and CLAHE produce grayscale output that would require channel conversion before detection.

5.2 Vehicle Detection Results

YOLOv8n was run on all 200 rain_storm images at confidence threshold 0.25. A total of 1,098 detections were recorded across the full dataset. Detection was also run on a gamma-corrected sample to measure enhancement impact.

Metric	Value	Notes
Total Images Processed	200	Full DAWN Rain subset
Total Detections	1,098	Across all 200 images
Avg Detections per Image	5.49	1,098 / 200 images
Original Image Detections (sample)	6 (image 1)	Persons, cars, motorcycle detected
Enhanced Image Detections	4	After Gamma Correction (γ=1.5): 3 cars, 1 truck

Table 3: Vehicle Detection Results Summary

Sample detection outputs for the first 5 images:

- Image 1: 6 detections (2 persons, 3 cars, 1 umbrella)
- Image 2: 5 detections (3 persons, 1 car, 1 motorcycle)
- Image 3: 1 detection (1 boat — misclassification due to rain spray)
- Image 4: 6 detections
- Image 5: 3 detections

The enhanced image (gamma correction applied) produced 4 detections: 3 cars and 1 truck, compared to variable results on the original. The gamma-corrected image detection output of "3 cars, 1 truck" reflects a more vehicle-focused detection pattern, with the brightness enhancement helping YOLOv8 distinguish vehicle silhouettes from rain streaks and wet road reflections.

5.3 GitHub Repository

All detection results were saved to the GitHub repository under the results/ folder using shutil.copytree(). The repository is publicly accessible at:

https://github.com/Nouman14-ai/CV_A2_LowVis_Nouman_22F-3788

6. Comparative Analysis

6.1 Enhancement Technique Comparison

The three enhancement techniques each offer different trade-offs. Histogram Equalisation produces the highest contrast output but loses colour information and can introduce over-enhancement artifacts in already bright regions (such as headlights and water spray). CLAHE provides better local adaptation and is less prone to over-enhancement, but still outputs grayscale. Gamma Correction is

the most detection-friendly because it maintains colour channels, preserves natural appearance, and brightens underexposed areas without introducing artifacts.

For integration with the YOLOv8 detection pipeline (pre-trained on colour COCO images), Gamma Correction is recommended as the primary enhancement method. CLAHE would be more suitable if a grayscale-trained detector or additional colour reconstruction step were used.

6.2 Comparison with Selected Paper (Raza et al., 2024)

The selected paper reports YOLOv5l at 77.30% mAP50 on the full DAWN dataset with models fine-tuned on DAWN annotations. This assignment uses YOLOv8n pre-trained on COCO without fine-tuning on DAWN, which explains the different detection patterns (COCO classes like person, umbrella, boat appearing in addition to vehicles). The reference benchmark of 77.30% mAP50 represents the target for a fully fine-tuned implementation.

The average of 5.49 detections per image reflects the dense traffic and complex scenes in the rain subset. Fine-tuning YOLOv8 on the DAWN-annotated rain images (4 vehicle classes: cars, buses, trucks, motorcycles) would produce class-specific mAP metrics directly comparable to the paper's results.

6.3 Impact of Enhancement on Detection

The gamma-corrected sample image yielded 4 detections (3 cars, 1 truck) — all vehicle classes — compared to a mixed detection of vehicles and non-vehicle objects on the original image. This suggests that brightness enhancement helps the pre-trained COCO model focus on vehicle-like features by reducing the rain streak and glare artifacts that may trigger non-vehicle class activations. This aligns with the reference paper's motivation for pre-processing before detection.

7. Challenges and Solutions

7.1 Non-Vehicle Class Detections

YOLOv8 pre-trained on COCO detected non-vehicle classes including umbrella, boat, and person — classes not relevant to the vehicle detection task. Image 3 produced only 1 detection (boat) due to rain spray patterns resembling water objects. This was addressed by noting it as a known limitation of using a non-domain-specific model, and is resolved in the reference paper's approach by fine-tuning YOLO specifically on DAWN vehicle classes.

7.2 Grayscale Output from Enhancement

Histogram Equalisation and CLAHE produce grayscale output, which is incompatible with the RGB-input YOLOv8 model without conversion. Gamma Correction was selected as the primary enhancement technique for the detection pipeline because it preserves colour channels. For future work, colour histogram equalisation (applied per-channel or in LAB colour space) would allow all three techniques to be tested with detection.

7.3 Dataset Access and Path Management

The dataset was accessed via GitHub clone rather than direct file upload, which introduced dependency on internet connectivity in the Colab environment. The path structure (dataset/Rain/)

was verified before loading. 2 images failed to load (200 loaded out of 202 total) due to potential file corruption, handled gracefully with the None check in the loading loop.

7.4 Lack of Ground Truth for mAP Calculation

The DAWN dataset labels are in YOLO format but were not directly loaded for mAP evaluation in this implementation. Full mAP calculation requires matching predicted bounding boxes to ground truth annotations with IoU thresholds. This is the primary step needed to reproduce the reference paper's 77.30% mAP50 benchmark.

8. Conclusion and Future Work

8.1 Summary of Achievements

- Successfully loaded and analysed 200 rain-storm images from the DAWN dataset with EDA including resolution and brightness distribution analysis.
- Implemented three image enhancement techniques: Histogram Equalisation, CLAHE (clipLimit=2.0), and Gamma Correction (γ=1.5) with visual comparison results.
- Applied YOLOv8n detection on all 200 images, achieving 1,098 total detections (average 5.49 per image).
- Demonstrated enhancement impact by comparing detection on original vs gamma-corrected images (4 vehicle-class detections on enhanced image).
- Saved all detection results to GitHub repository and documented the complete pipeline.

8.2 Key Findings

- Gamma Correction (γ=1.5) is the most suitable enhancement for colour-based detection pipelines, preserving image channels while brightening underexposed rainy scenes.
- Pre-trained YOLOv8n on COCO achieves reasonable detection rates (avg 5.49 per image) on rain-storm images without domain-specific fine-tuning.
- Non-vehicle class false detections (umbrella, boat) highlight the importance of domain-specific fine-tuning for accurate mAP evaluation.
- Brightness enhancement helps reduce rain artifact false positives and improves vehicle-class detection focus.

8.3 Limitations

- YOLOv8 was not fine-tuned on DAWN, so mAP50 cannot be compared to the reference paper's 77.30% benchmark.
- Only the rain subset was used; performance on fog, haze, and snow subsets was not evaluated.
- Enhancement was applied only to sample images for detection comparison; full dataset enhancement and re-detection was not performed.

8.4 Future Work

- Fine-tune YOLOv8 on DAWN-annotated rain images (4 classes) to compute mAP50 and compare directly with Raza et al. (2024).
- Implement attention-enhanced YOLOv5s variants (CBAM, NAM, SimAM) from the selected paper to replicate its detection architecture.

- Extend enhancement to all 200 images and run detection on enhanced dataset to compute aggregate improvement metrics.
- Apply colour CLAHE (in LAB colour space) to enable all three enhancement techniques to be used with the colour YOLOv8 pipeline.
- Implement the Deep-SORT tracking component from the reference paper for multi-vehicle tracking across video frames.
- Evaluate on fog, haze, and snow subsets for cross-weather generalisation analysis.

9. References

[1] N. Raza, M. A. Habib, M. Ahmad, Q. Abbas, M. B. Aldajani, and M. A. Latif, "Efficient and Cost-Effective Vehicle Detection in Foggy Weather for Edge/Fog-Enabled Traffic Surveillance and Collision Avoidance Systems," Computers, Materials & Continua, vol. 81, no. 1, pp. 911–931, 2024. https://ieeexplore.ieee.org/abstract/document/11078259

[2] J. Gao, L. Dai, W. Zhang, Q. Hu, G. Chen, and Z. Chen, "Pedestrian and Vehicle Detection in Foggy Weather Using DehazeFormer," The Visual Computer, Springer, vol. 41, pp. 8523–8535, 2025. https://doi.org/10.1007/s00371-025-03882-0

[3] K. He, J. Sun, and X. Tang, "Single Image Haze Removal Using Dark Channel Prior," IEEE Trans. Pattern Anal. Mach. Intell., vol. 33, no. 12, pp. 2341–2353, 2011. https://ieeexplore.ieee.org/document/5567108/

[4] M. Hassaballah, M. A. Kenk, K. Muhammad, and S. Minaee, "Vehicle Detection and Tracking in Adverse Weather Using a Deep Learning Framework," IEEE Trans. Intell. Transp. Syst., vol. 22, no. 7, pp. 4230–4242, 2021.

[5] M. A. Kenk and M. Hassaballah, "DAWN: Vehicle Detection in Adverse Weather Nature Dataset," arXiv preprint arXiv:2008.05402, 2020. https://arxiv.org/abs/2008.05402

[6] Ultralytics, "YOLOv8: A New State-of-the-Art Computer Vision Model," 2023. https://docs.ultralytics.com

[7] DAWN Dataset — Kaggle: https://www.kaggle.com/datasets/shuvoalok/dawn-dataset

[8] GitHub Repository: https://github.com/Nouman14-ai/CV_A2_LowVis_Nouman_22F-3788